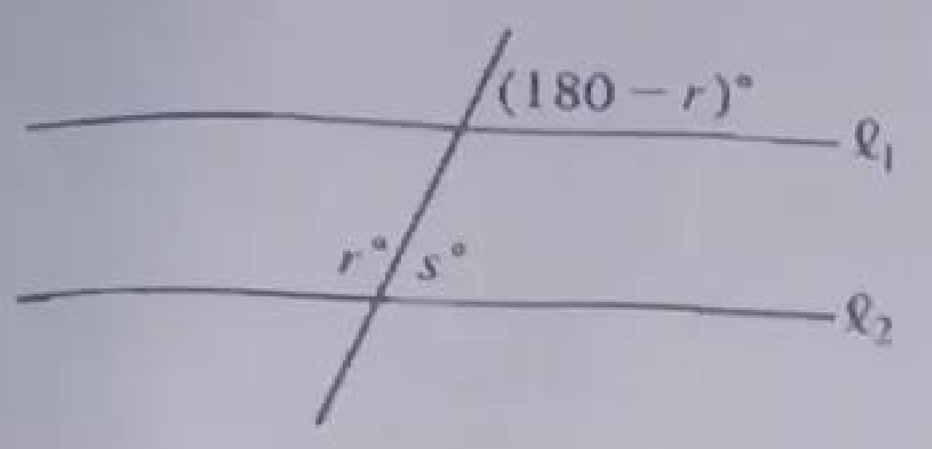
Course Title: Mathematical Analysis for Computer Science

Course Code: CSE-3201

1. a)

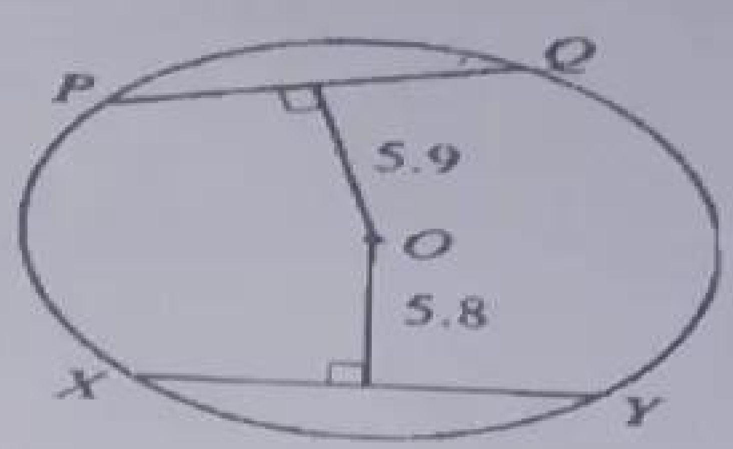


$\ell_1 \parallel \ell_2$

s

60

b)

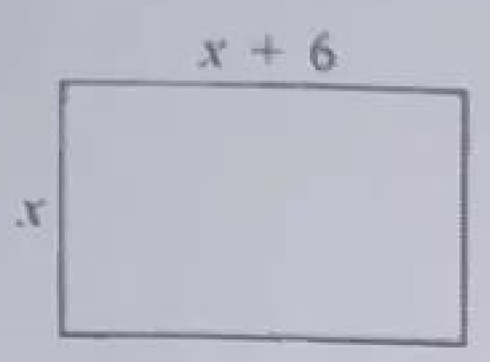


Circle with center O

The length of chord PQ

The length of chord XY

c)

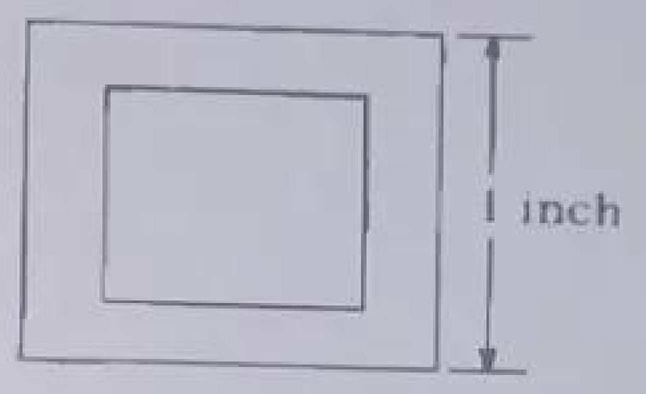


The perimeter of square S is equal to the perimeter of the rectangle above.

The length of a side of S

$x + 3$

d)



In the figure above, if the area of the smaller square region is $\frac{1}{2}$ the area of the larger square region, then the diagonal of the larger square is how many inches longer than the diagonal of the smaller square?

e)

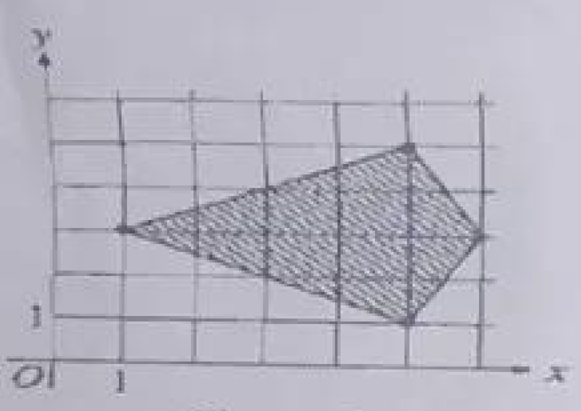


Figure 1

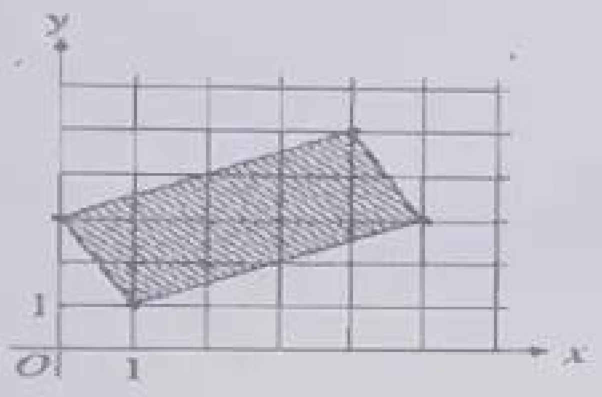


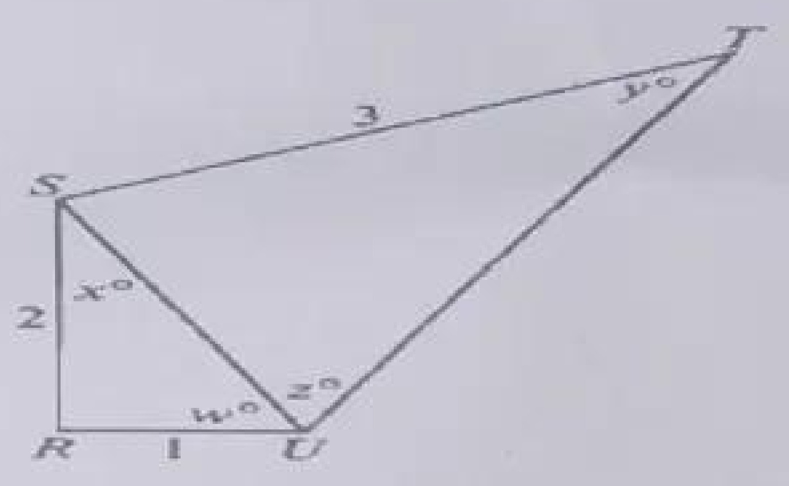
Figure 2

Note: Drawn to scale.

The area of the shaded region shown in Figure 1

The area of the shaded region shown in Figure 2

f)



$w + x = 90$
$y + z = 90$

The perimeter of $RSTU$

10

g)

Mary has 3 dollars more than Bill has, but 5 dollars less than Jane has. If Mary has x dollars, how many dollars do Jane and Bill have altogether?

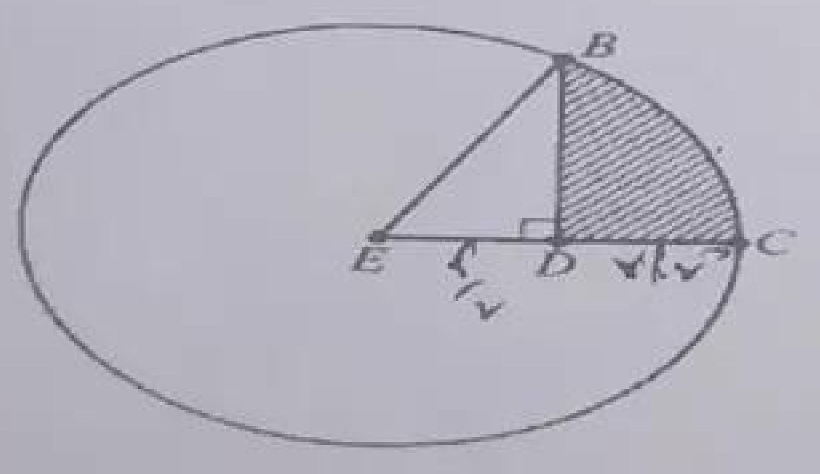
h)

$d > 0$

The total interest earned on d dollars invested for 3 months at 11 percent simple annual interest

$\frac{11}{3}\left(\frac{d}{100}\right)$ dollars

i)



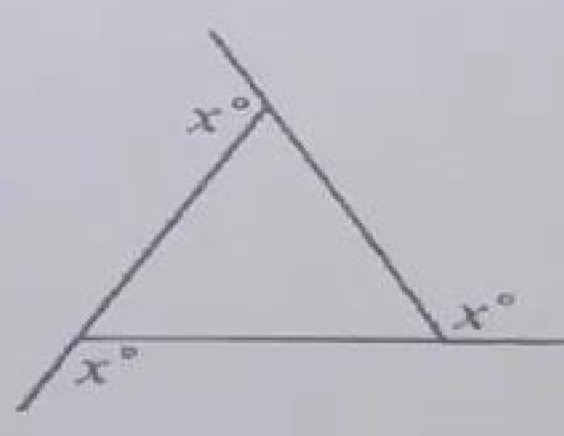
The circle with center E has radius r.

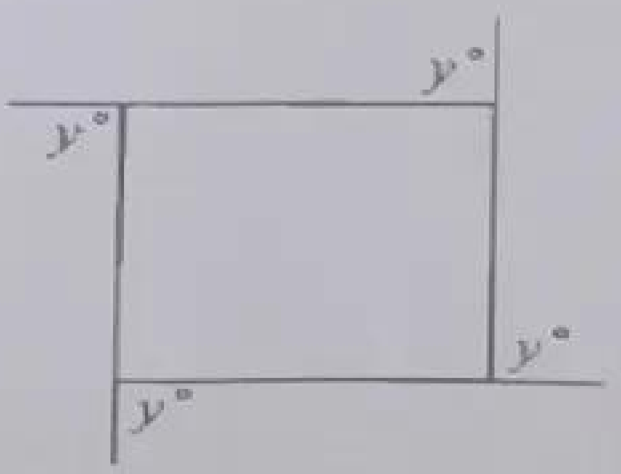
$ED = \frac{r}{2}$

The area of $\triangle EBD$

The area of the shaded region

j)





$x - y$

$\frac{x + y}{7}$

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